

STT 861: Theory of Probability & Statistics I

Lecture 9: Transformations of Random Variables I

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Reference: Casella & Berger Ch. 2.1

Motivation — Why Transform Random Variables?

Setting

If X is a random variable with cdf $F_X(x)$, then any function $Y = g(X)$ is also a random variable. We want to find the distribution of Y .

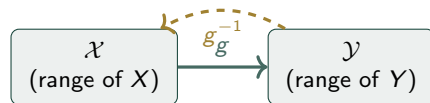
Key idea: For any set A ,

$$P(Y \in A) = P(g(X) \in A) = P(X \in g^{-1}(A))$$

where the **inverse image** is

$$g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}$$

This links the distribution of Y back to that of X .



CDF approach (2.1.4)

For any y :

$$F_Y(y) = P(Y \leq y) = \int_{\{x: g(x) \leq y\}} f_X(x) dx$$

Transformations of Discrete Random Variables

PMF of $Y = g(X)$ (Discrete Case)

If X is discrete with pmf $f_X(x)$, then $Y = g(X)$ is also discrete with

$$f_Y(y) = P(Y = y) = \sum_{x: g(x)=y} f_X(x) = \sum_{x \in g^{-1}(y)} f_X(x), \quad y \in \mathcal{Y}$$

and $f_Y(y) = 0$ for $y \notin \mathcal{Y}$.

Example 2.1.1 (CB): Binomial Transformation

$X \sim \text{Binomial}(n, p)$: $f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, \dots, n$.

Let $Y = g(X) = n - X$. Then $g^{-1}(y) = \{n - y\}$, so:

$$\begin{aligned} f_Y(y) &= f_X(n - y) = \binom{n}{n - y} p^{n-y} (1-p)^{n-(n-y)} \\ &= \binom{n}{y} (1-p)^y p^{n-y} \end{aligned}$$

$\therefore Y \sim \text{Binomial}(n, 1 - p)$. (*Swapping successes and failures swaps p and $1 - p$.*)

The CDF Method — General Approach

CDF Method (Equation 2.1.4)

For $Y = g(X)$ with X continuous:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \int_{\{x \in \mathcal{X}: g(x) \leq y\}} f_X(x) dx$$

Once $F_Y(y)$ is found, differentiate to get $f_Y(y) = F'_Y(y)$.

Example 2.1.2 (CB): Uniform $\rightarrow \sin^2$ Transformation

$X \sim \text{Uniform}(0, 2\pi)$, so $f_X(x) = \frac{1}{2\pi}$ on $(0, 2\pi)$.

Let $Y = \sin^2(X)$. For $0 < y < 1$, the set $\{x : g(x) \leq y\}$ has four pieces (from the graph of $\sin^2$):

$$P(Y \leq y) = P(X \leq x_1) + P(x_2 \leq X \leq x_3) + P(X \geq x_4)$$

Using symmetry of $\sin^2$ and $X \sim \text{Uniform}$:

$$F_Y(y) = 2P(X \leq x_1) + 2P(x_2 \leq X \leq \pi) = \frac{2x_1}{2\pi} + \frac{2(\pi - x_2)}{2\pi}$$

where x_1, x_2 solve $\sin^2(x) = y$ on $(0, \pi)$. The resulting F_Y is not simple — illustrating the difficulty of non-monotone transformations.

Support Sets and Monotone Functions

Support (Equation 2.1.7)

For the transformation $Y = g(X)$, it is convenient to work with:

$$\mathcal{X} = \{x : f_X(x) > 0\} \quad \text{and} \quad \mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}$$

$\mathcal{X}$ is the **support** of X ; $\mathcal{Y}$ is the induced sample space of Y .

Monotone functions are one-to-one

- g is **increasing**: $u > v \Rightarrow g(u) > g(v)$.
- g is **decreasing**: $u < v \Rightarrow g(u) > g(v)$.
- If g is monotone on $\mathcal{X}$, then $g^{-1}(y)$ is a single point for each $y \in \mathcal{Y}$.
- Increasing: $\{x : g(x) \leq y\} = \{x : x \leq g^{-1}(y)\}$
- Decreasing: $\{x : g(x) \leq y\} = \{x : x \geq g^{-1}(y)\}$

Theorem 2.1.3 — CDF Under Monotone Transformation

Theorem 2.1.3 (CB)

Let X have cdf $F_X(x)$, let $Y = g(X)$, and let $\mathcal{X}, \mathcal{Y}$ be as in (2.1.7).

- Ⓐ If g is **increasing** on $\mathcal{X}$:

$$F_Y(y) = F_X(g^{-1}(y)), \quad y \in \mathcal{Y}$$

- Ⓑ If g is **decreasing** on $\mathcal{X}$ and X is continuous:

$$F_Y(y) = 1 - F_X(g^{-1}(y)), \quad y \in \mathcal{Y}$$

Example 2.1.4 (CB): Uniform-Exponential Relationship

$X \sim \text{Uniform}(0, 1)$: $f_X(x) = 1$ on $(0, 1)$, $F_X(x) = x$.

Let $Y = g(X) = -\log X$. Since $g'(x) = -1/x < 0$, g is **decreasing**.

$g^{-1}(y) = e^{-y}$, $\mathcal{Y} = (0, \infty)$.

By Theorem 2.1.3(b): $F_Y(y) = 1 - F_X(e^{-y}) = 1 - e^{-y}$, $y > 0$.

$\therefore Y \sim \text{Exponential}(1)$. *This is the probability integral transform — a fundamental result!*

Theorem 2.1.5 — PDF Under Monotone Transformation

Theorem 2.1.5 (CB) — Change of Variables

Let X have pdf $f_X(x)$, let $Y = g(X)$ where g is **monotone** on $\mathcal{X}$, and suppose $g^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$. Then:

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| & y \in \mathcal{Y} \\ 0 & \text{otherwise} \end{cases}$$

Proof: Differentiate Theorem 2.1.3 using the chain rule. $\square$

How to apply Thm 2.1.5

1. Verify g is monotone on $\mathcal{X}$.
2. Find the inverse $x = g^{-1}(y)$ and its derivative $\frac{d}{dy}g^{-1}(y)$.
3. Substitute into the formula: $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy}g^{-1}(y) \right|$.
4. State the support $\mathcal{Y}$.

Example 2.1.6 — Inverted Gamma PDF

Example 2.1.6 (CB): Inverted Gamma

Let X have the **gamma pdf**:

$$f_X(x) = \frac{1}{(n-1)! \beta^n} x^{n-1} e^{-x/\beta}, \quad 0 < x < \infty$$

Let $Y = g(X) = 1/X$. Then $g'(x) = -1/x^2 < 0$ (decreasing on $(0, \infty)$).

$$g^{-1}(y) = 1/y, \quad \frac{d}{dy} g^{-1}(y) = -1/y^2.$$

By Theorem 2.1.5:

$$\begin{aligned} f_Y(y) &= f_X(1/y) \cdot |-1/y^2| = \frac{1}{(n-1)! \beta^n} \left(\frac{1}{y}\right)^{n-1} e^{-1/(\beta y)} \cdot \frac{1}{y^2} \\ &= \frac{1}{(n-1)! \beta^n} \left(\frac{1}{y}\right)^{n+1} e^{-1/(\beta y)}, \quad y > 0. \text{ This is the } \mathbf{inverted \text{ gamma}} \text{ distribution.} \end{aligned}$$

Example 2.1.7 — Square Transformation (Non-Monotone)

Example 2.1.7 (CB)

X continuous, $Y = X^2$.

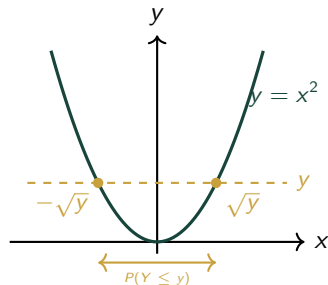
$g(x) = x^2$ is **not** monotone on $\mathbb{R}$: decreasing on $(-\infty, 0)$, increasing on $(0, \infty)$.

CDF approach: For $y > 0$:

$$\begin{aligned}F_Y(y) &= P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\&= F_X(\sqrt{y}) - F_X(-\sqrt{y})\end{aligned}$$

Differentiate:

$$f_Y(y) = \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})]$$



Theorem 2.1.8 — Piecewise Monotone Transformations

Theorem 2.1.8 (CB) — General Formula

Suppose $\mathcal{X}$ can be partitioned into sets $A_0, A_1, \dots, A_k$ with $P(X \in A_0) = 0$, and on each A_i there exists a monotone function $g_i(x) = g(x)$ with continuous inverse $g_i^{-1}(y)$ on $\mathcal{Y}$. Then:

$$f_Y(y) = \begin{cases} \sum_{i=1}^k f_X(g_i^{-1}(y)) \left| \frac{d}{dy} g_i^{-1}(y) \right| & y \in \mathcal{Y} \\ 0 & \text{otherwise} \end{cases}$$

Key point

The support $\mathcal{X}$ is split into pieces where g is monotone. Each piece contributes one term to the sum. The “exceptional set” A_0 has probability zero and can be ignored.

Example 2.1.9 — Normal to Chi-Squared

Example 2.1.9 (CB): $X \sim N(0, 1) \Rightarrow Y = X^2 \sim \chi_1^2$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

$g(x) = x^2$: monotone on $A_1 = (-\infty, 0)$ and $A_2 = (0, \infty)$.

$A_0 = \{0\}$ has probability zero. $\mathcal{Y} = (0, \infty)$.

$$g_1^{-1}(y) = -\sqrt{y}, \quad \frac{d}{dy} g_1^{-1}(y) = \frac{-1}{2\sqrt{y}}$$

$$g_2^{-1}(y) = \sqrt{y}, \quad \frac{d}{dy} g_2^{-1}(y) = \frac{1}{2\sqrt{y}}$$

By Theorem 2.1.8:

$$\begin{aligned} f_Y(y) &= f_X(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} \\ &= \frac{2}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-y/2}, \quad y > 0 \end{aligned}$$

This is the **chi-squared distribution with 1 degree of freedom**.

Section 2.1 — Summary

Case	Method	Formula
Discrete X , any g	Sum over $g^{-1}(y)$	$f_Y(y) = \sum_{x: g(x)=y} f_X(x)$
Continuous, any g	CDF method	$F_Y(y) = \int_{\{g(x) \leq y\}} f_X(x) dx$
Monotone g	Thm 2.1.3 (cdf) Thm 2.1.5 (pdf)	$f_Y(y) = f_X(g^{-1}(y)) \left \frac{d}{dy} g^{-1}(y) \right $
Piecewise monotone	Thm 2.1.8	$f_Y(y) = \sum_{i=1}^k f_X(g_i^{-1}(y)) \left \frac{d}{dy} g_i^{-1}(y) \right $

Key Examples

- **Ex 2.1.1:** $\text{Binomial}(n, p) \xrightarrow{Y=n-X} \text{Binomial}(n, 1-p)$
- **Ex 2.1.4:** $\text{Uniform}(0, 1) \xrightarrow{Y=-\log X} \text{Exponential}(1)$ (*probability integral transform*)
- **Ex 2.1.6:** $\text{Gamma} \xrightarrow{Y=1/X} \text{Inverted Gamma}$
- **Ex 2.1.9:** $N(0, 1) \xrightarrow{Y=X^2} \chi_1^2$

Example A — Linear Transformation

Example A: $Y = aX + b$

Let X have pdf $f_X(x)$ with support $\mathcal{X}$. Let $Y = aX + b$ where $a \neq 0$.

Step 1. $g(x) = ax + b$ is monotone ($g'(x) = a$).

Step 2. Inverse: $g^{-1}(y) = \frac{y-b}{a}$, $\frac{d}{dy}g^{-1}(y) = \frac{1}{a}$.

Step 3. By Theorem 2.1.5:

$$f_Y(y) = f_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{|a|}, \quad y \in \mathcal{Y} = \{ax + b : x \in \mathcal{X}\}$$

Specialisation: Standardising a Normal

Let $X \sim N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$.

Set $Y = \frac{X - \mu}{\sigma}$ (i.e. $a = 1/\sigma$, $b = -\mu/\sigma$). Then:

$$f_Y(y) = f_X(\sigma y + \mu) \cdot \sigma = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

$\therefore Y \sim N(0, 1)$. *Standardising a normal gives a standard normal.*

Example B — Exponential to Weibull

Example B: Power Transformation of an Exponential

Let $X \sim \text{Exponential}(1)$: $f_X(x) = e^{-x}$, $x > 0$.

Let $Y = X^{1/\beta}$ for $\beta > 0$.

Step 1. $g(x) = x^{1/\beta}$ is increasing on $(0, \infty)$. $\mathcal{Y} = (0, \infty)$.

Step 2. $g^{-1}(y) = y^\beta$, $\frac{d}{dy}g^{-1}(y) = \beta y^{\beta-1}$.

Step 3. By Theorem 2.1.5:

$$f_Y(y) = f_X(y^\beta) \cdot \beta y^{\beta-1} = e^{-y^\beta} \cdot \beta y^{\beta-1}, \quad y > 0$$

This is the Weibull Distribution

The pdf $f_Y(y) = \beta y^{\beta-1} e^{-y^\beta}$ for $y > 0$ is the **Weibull**(β) distribution, widely used in reliability and survival analysis. When $\beta = 1$ it reduces to Exponential(1).

Example C — Uniform to General Uniform

Example C: $\text{Uniform}(0, 1) \rightarrow \text{Uniform}(a, b)$

Let $X \sim \text{Uniform}(0, 1)$: $f_X(x) = 1$, $0 < x < 1$.

Let $Y = a + (b - a)X$ for $a < b$.

Step 1. $g(x) = a + (b - a)x$ is increasing. $\mathcal{Y} = (a, b)$.

Step 2. $g^{-1}(y) = \frac{y - a}{b - a}$, $\frac{d}{dy}g^{-1}(y) = \frac{1}{b - a}$.

Step 3.

$$f_Y(y) = f_X\left(\frac{y - a}{b - a}\right) \cdot \frac{1}{b - a} = 1 \cdot \frac{1}{b - a} = \frac{1}{b - a}, \quad a < y < b$$

Simulation from Uniform

$\therefore Y \sim \text{Uniform}(a, b)$.

This is the **standard simulation trick**: if $U \sim \text{Uniform}(0, 1)$, then $a + (b - a)U \sim \text{Uniform}(a, b)$.
More generally, if F is a cdf, then $F^{-1}(U)$ has cdf F — the *inverse probability integral transform*.

Example D — Absolute Value Transformation

Example D: $Y = |X|$ where X is symmetric

Let X have pdf $f_X(x)$ symmetric about 0 (i.e. $f_X(-x) = f_X(x)$).

Let $Y = |X|$. Then for $y > 0$:

$$F_Y(y) = P(|X| \leq y) = P(-y \leq X \leq y) = F_X(y) - F_X(-y)$$

Differentiating:

$$f_Y(y) = f_X(y) + f_X(-y) = 2f_X(y), \quad y > 0$$

(using symmetry $f_X(-y) = f_X(y)$).

Specialisation: Folded Normal

If $X \sim N(0, 1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.

Then $Y = |X|$ has pdf:

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \sqrt{\frac{2}{\pi}} e^{-y^2/2}, \quad y > 0$$

This is the **half-normal** (or folded normal) distribution.

Example E — Log-Normal Distribution

Example E: $Y = e^X$ where $X \sim N(\mu, \sigma^2)$

Let $X \sim N(\mu, \sigma^2)$ with pdf $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$.

Let $Y = e^X$ (equivalently $X = \log Y$).

Step 1. $g(x) = e^x$ is increasing. $\mathcal{Y} = (0, \infty)$.

Step 2. $g^{-1}(y) = \log y$, $\frac{d}{dy}g^{-1}(y) = \frac{1}{y}$.

Step 3.

$$f_Y(y) = f_X(\log y) \cdot \frac{1}{y} = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log y - \mu)^2}{2\sigma^2}\right\}, \quad y > 0$$

Log-Normal Distribution

$Y = e^X \sim \text{LogNormal}(\mu, \sigma^2)$. Used to model stock prices, incomes, lifetimes — quantities that are always positive and whose *logarithm* is normally distributed.

In-Class Activity

Q1 — Monotone Transformation (Increasing)

Let $X \sim \text{Exponential}(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$, $x > 0$.

Let $Y = 2X + 3$.

(a) Find $g^{-1}(y)$ and $\frac{d}{dy}g^{-1}(y)$. **(b)** Find $f_Y(y)$ and its support. **(c)** What is $E[Y]$ (use $E[X] = 1/\lambda$)?

Q2 — Monotone Transformation (Decreasing)

Let $X \sim \text{Uniform}(0, 1)$: $f_X(x) = 1$, $0 < x < 1$.

Let $Y = -\log X$.

(a) Is $g(x) = -\log x$ increasing or decreasing?

(b) Find $F_Y(y)$ using Theorem 2.1.3. **(c)** Find $f_Y(y)$. What distribution is Y ?

Q3 — Non-Monotone Transformation

Let $X \sim \text{Uniform}(-1, 1)$: $f_X(x) = \frac{1}{2}$, $-1 < x < 1$.

Let $Y = X^2$.

(a) Find $F_Y(y) = P(X^2 \leq y)$ for $0 < y < 1$.

(b) Find $f_Y(y)$ by differentiating. (c) Verify $\int_0^1 f_Y(y) dy = 1$.

Q4 — Identify the Distribution

Let $X \sim \text{Exponential}(1)$ and $Y = 1 - e^{-X}$.

(a) Find $g^{-1}(y)$ and its derivative. (b) Find $f_Y(y)$. (c) What well-known distribution is Y ?

Activity Solutions — Q1 & Q2

Q1 Solution: $Y = 2X + 3$, $X \sim \text{Exp}(\lambda)$

(a) $g^{-1}(y) = \frac{y-3}{2}$; $\frac{d}{dy}g^{-1}(y) = \frac{1}{2}$.

(b) Support: $y > 3$ (since $x > 0 \Rightarrow y = 2x + 3 > 3$).

$$f_Y(y) = f_X\left(\frac{y-3}{2}\right) \cdot \frac{1}{2} = \lambda e^{-\lambda(y-3)/2} \cdot \frac{1}{2} = \frac{\lambda}{2} e^{-\frac{\lambda}{2}(y-3)}, \quad y > 3$$

$Y - 3 \sim \text{Exp}(\lambda/2)$, i.e. $Y \sim 3 + \text{Exp}(\lambda/2)$.

(c) $E[Y] = 2E[X] + 3 = \frac{2}{\lambda} + 3$.

Q2 Solution: $Y = -\log X$, $X \sim \text{Uniform}(0, 1)$

(a) $g'(x) = -1/x < 0$ for $x \in (0, 1)$: **decreasing**. $\mathcal{Y} = (0, \infty)$.

(b) $g^{-1}(y) = e^{-y}$. By Theorem 2.1.3(b): $F_Y(y) = 1 - F_X(e^{-y}) = 1 - e^{-y}$, $y > 0$.

(c) $f_Y(y) = F'_Y(y) = e^{-y}$, $y > 0$. This is Exponential(1).

(Confirms Example 2.1.4 from CB — the probability integral transform.)

Activity Solutions — Q3 & Q4

Q3 Solution: $Y = X^2$, $X \sim \text{Uniform}(-1, 1)$

(a) For $0 < y < 1$:

$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \frac{2\sqrt{y}}{2} = \sqrt{y}$$

(b) $f_Y(y) = \frac{d}{dy} \sqrt{y} = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$

(c) $\int_0^1 \frac{1}{2\sqrt{y}} dy = [\sqrt{y}]_0^1 = 1. \checkmark$

Note: $f_Y(y) = \frac{1}{2}y^{-1/2} = \frac{1}{2}y^{1/2-1}$ is a $\text{Beta}(\frac{1}{2}, 1)$ pdf (up to normalisation).

Q4 Solution: $Y = 1 - e^{-X}$, $X \sim \text{Exp}(1)$

(a) $y = 1 - e^{-x} \Rightarrow e^{-x} = 1 - y \Rightarrow g^{-1}(y) = -\log(1 - y)$; $\frac{d}{dy}g^{-1}(y) = \frac{1}{1 - y}$.

(b) Support: $y \in (0, 1)$ (since $x > 0 \Rightarrow e^{-x} \in (0, 1) \Rightarrow y \in (0, 1)$).

$$f_Y(y) = f_X(-\log(1 - y)) \cdot \frac{1}{1 - y} = e^{-(-\log(1 - y))} \cdot \frac{1}{1 - y} = (1 - y) \cdot \frac{1}{1 - y} = 1, \quad 0 < y < 1$$

(c) $f_Y(y) = 1$ on $(0, 1)$: $Y \sim \text{Uniform}(0, 1)$!

*This is the **probability integral transform**: if $X \sim F$, then $F(X) \sim \text{Uniform}(0, 1)$.*